

Kishan K

Software Engineer | AI and ML Engineer

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GitHub: github.com/kishanaik5 | LeetCode: leetcode.com/u/ZXO2yYfahk | Portfolio: kishan-k.pages.dev

EDUCATION

Ramaiah Institute of Technology, Bengaluru
B.E. in Artificial Intelligence and Machine Learning

Dec 2021 – Jul 2025
CGPA: 8.6 / 10

EXPERIENCE

Sumedha Design Systems Pvt. Ltd. – AI and Automation Intern Hyderabad | Feb 2025 – Aug 2025

- Streamlined a multimodal data extraction pipeline using **GCP Vision API** and **AWS S3** to process **10,000+** pages of VLSI documentation and PDFs; engineered automated **RAG** systems for MCQ generation that reduced manual content creation time by **60%** through intelligent document chunking and metadata-driven retrieval.
- Spearheaded multi-agent LLM workflows and architected an advanced retrieval framework integrating **LlamaIndex** and **FAISS** for vector similarity search; implemented hybrid search and re-ranking algorithms that boosted learner engagement metrics by **15%** and reduced system latency by **50%** during high-concurrency student usage.

Silo Fortune Private Limited – Software Engineer (AI/ML) Bengaluru | Oct 2025 – Present

- Designed and deployed the **Kisaan Sampurna** multimodal intelligence pipeline, merging computer vision disease detection with **Gemini 2.5 Flash** for real-time treatment advisory; integrated satellite (NDVI/UVI) and 7-day weather APIs to produce multilingual advisories for **12+** Indian languages, achieving **92%+** precision.
- Orchestrated a production-grade **FastAPI microservices ecosystem** with PostgreSQL isolation, **Redis** caching, and Docker CI/CD, improving reliability and cutting downtime by **40%**; built a secure AI orchestration layer for plugin-based routing across ERP and payment workflows, handling over **10,000+** API requests daily.
- Built **Real-Time Mandi Price API** serving live price data for **7,000+** agricultural markets with nearest-mandi auto-resolution using Haversine formula in SQL, multilingual support across 12 languages, and dual-source data architecture (PostgreSQL + AWS S3 Parquet) with cascade dropdowns under 5ms.
- Engineered **AI Farm Intelligence System** converting satellite indices (NDVI, NRI, EVI, DSWI) into 432 farm condition fingerprints via deterministic bucketing, generating 15-section LLM advisories in **13 Indian languages** through a two-level cache driving LLM cost to zero deployed on **AWS ECS Fargate** with Jenkins CI/CD.
- Built **AI Crop Stage Prediction** pipeline consuming 5 years of satellite imagery to detect crop growth cycles for **21 crops** across **80+ sub-stages** using VegetationIntelligenceEngine (VIE); generated pest/disease advisories via parallel `asyncio.gather` Gemini calls delivered in 10 Indian languages via cache-first translation.

SKILLS

Programming:	Python (AsyncIO) C/C++ SQL
AI & GenAI:	Google Gemini (Flash/Pro) OpenAI API TensorFlow PyTorch Transformers ViT CNNs
MLOps & DevOps:	Docker/Compose GitHub Actions CI/CD MLflow DVC AWS SageMaker Jenkins
Web/Backend:	FastAPI Django Flask Pydantic SQLAlchemy Alembic React.js React Native Node.js REST APIs
Cloud Platforms:	AWS (RDS/S3/ECS Fargate) GCP Azure Render Vercel
AI Frameworks:	LangChain LlamaIndex Rasa Dialogflow Streamlit LiveKit
Databases:	PostgreSQL MongoDB MySQL FAISS (Vector DB) Redis

PROJECTS

Real-Time Mandi Price API GitHub: github.com/kishanaik5/mandi_price_forecast_predictor

- Architected a **FastAPI** backend serving real-time price data for **7,000+** agricultural markets (mandis) across India; implemented nearest-mandi auto-resolution using the **Haversine formula directly in SQL**, 12-language multilingual support, and dual-source data (PostgreSQL for hot queries, AWS S3 Parquet for bulk/historical) with cascade dropdowns responding in under **5ms**.
- Tech:* FastAPI, PostgreSQL, AWS S3, Docker, SQLAlchemy, Alembic, JWT

AI Crop Stage Prediction GitHub: github.com/kishanaik5/Crop_stages_prediction_model

- Built a 5-layer intelligence pipeline reading **5 years of satellite imagery** (NDVI, EVI, NDMI via AgroMonitoring API) through a custom VegetationIntelligenceEngine to detect growth cycles, map **80+ sub-stages** for **21 crops**,

and generate pest/disease advisories using **16 parallel async Gemini calls** per crop; delivered in 10 Indian languages via cache-first bulk translation.

- *Tech:* FastAPI, Google Gemini 2.0 Flash, AgroMonitoring API, PostgreSQL, AWS S3, asyncio

AI Farm Intelligence System

GitHub: github.com/kishanaik5/Yield-Weather-Forecast

- Engineered a satellite-to-advisory pipeline converting 6 spectral indices (NDVI, NRI, EVI, EVI2, NDWI, DSWI) into **432 farm condition fingerprints** via deterministic bucketing; Gemini generates **15-section advisory reports** in **13 Indian languages** via a two-level Knowledge Base cache (English template + translation), driving LLM cost to near-zero at scale (5,616 max cache entries); deployed on **AWS ECS Fargate** with Jenkins CI/CD.
- *Tech:* FastAPI, Google Gemini, AWS S3, AWS ECS Fargate, PostgreSQL (JSONB), Jenkins, AgroMonitoring API

Yield & Weather Forecast Agent

GitHub: [Yield-Weather-Forecast](https://github.com/kishanaik5/Yield-Weather-Forecast)

- Architected a **FastAPI** backend integrating Agromonitoring and Google Maps APIs to deliver localized 5-day forecasts; implemented a rule-based normalization engine and **IST date-based caching** in PostgreSQL to reduce API latency.
- *Tech:* FastAPI, PostgreSQL, Pydantic, Redis, Docker, Alembic, Agromonitoring API

Agri-Expert Agent

GitHub: tinyurl.com/github-agri-repo

- Devised an intelligent diagnostic system using deep learning and Knowledge-Based Agents to automate identification of **360+** crop varieties and diseases in **10** languages with **94%** precision.
- *Tech:* Python, Computer Vision, Deep Learning, ML Agents, Expert Systems

RAG for VLSI Automation

GitHub: tinyurl.com/github-RAG-VLSI

- Constructed an AI-powered RAG system using Transformers to automate generation of **2,000+** technical MCQs, achieving high contextual relevance and accelerating professional content creation by **3x**.
- *Tech:* Python, HuggingFace, MongoDB, AWS S3, Prompt Engineering

Waste Segregation IoT

GitHub: tinyurl.com/github-DL-Waste-CV-Model

- An IoT-based domestic waste segregation solution utilizing Inception ResNet V2, successfully achieving **92.5%** test accuracy across **6** distinct waste categories.
- *Tech:* PyTorch, ReactJS, Flask, ESP-32

PUBLICATIONS

Performance Evaluation of Deep Learning Models for Predicting Alzheimer's Disease

IEEE Bangalore Section, 2024

Enhancing Predictive Maintenance with SHAP and LIME: A Framework for Explainable AI

ICAI – ARSSS, 2025

CERTIFICATIONS

100 Days of Code: The Complete Python Pro Bootcamp

Udemy

Machine Learning, Data Science and Generative AI with Python

Udemy

Complete Data Science, Machine Learning, Deep Learning, NLP Bootcamp

Udemy

Introduction to Microsoft Azure Cloud Services

Coursera

AWS Cloud Support Associate

Coursera